

SET 23: Employee Performance Analysis (RStudio)

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1 Sales 5 85

2 HR 3 92

3 Marketing 7 78

4 Sales 4 90

5 HR 2 76

- 1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.**
- 2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.**
- 3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.**
- 4. In R, develop a table showing the performance data for each employee. Label the table elements.**

Dataset is created inside the R code (no CSV file required).

Step 1: Create the Dataset

```
employee <- data.frame(  
  Employee_ID = c(1,2,3,4,5),  
  Department = c("Sales","HR","Marketing","Sales","HR"),  
  Years_of_Service = c(5,3,7,4,2),  
  Performance_Score = c(85,92,78,90,76)  
)
```

```
View(employee)
```

Experiment 1: Bar Chart (Department Distribution)

Install Package (Only Once)

```
install.packages("ggplot2")
```

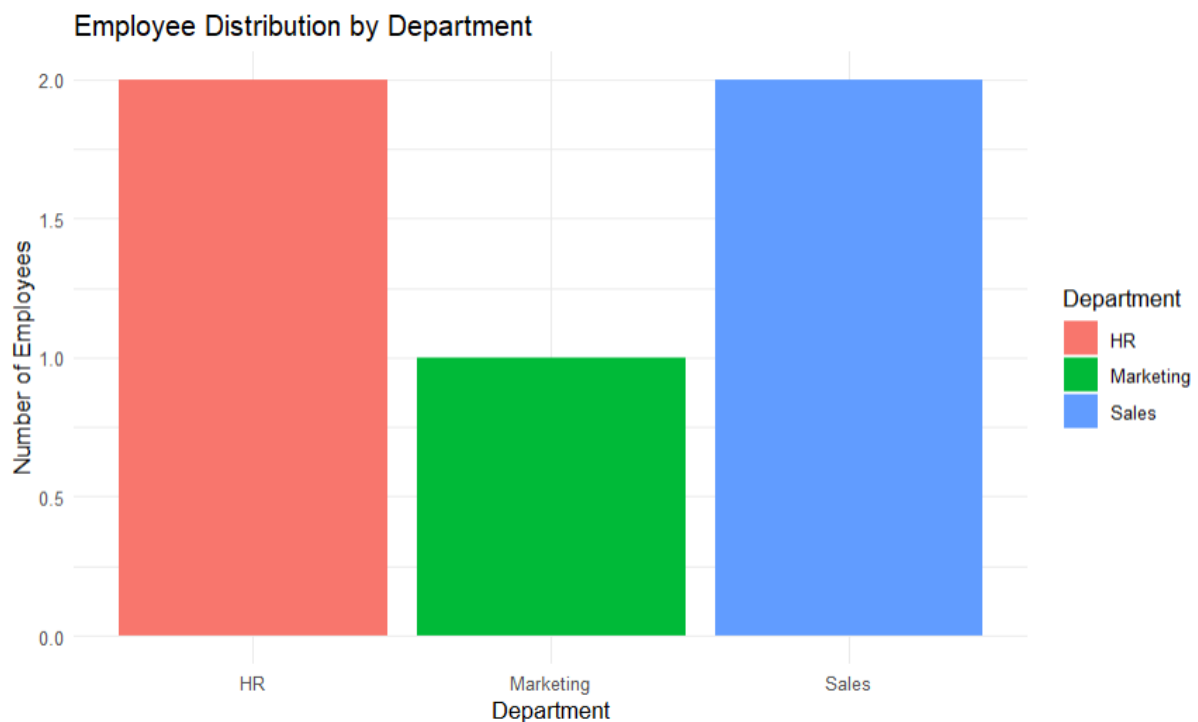
Load Package

```
library(ggplot2)
```

Bar Chart

```
ggplot(employee,  
  aes(x = Department,  
      fill = Department)) +  
geom_bar() +  
labs(  
  title = "Employee Distribution by Department",  
  x = "Department",  
  y = "Number of Employees"  
) +  
theme_minimal()
```

Output



A bar chart showing the number of employees in each department.

Experiment 2: Line Chart (Performance Trend)

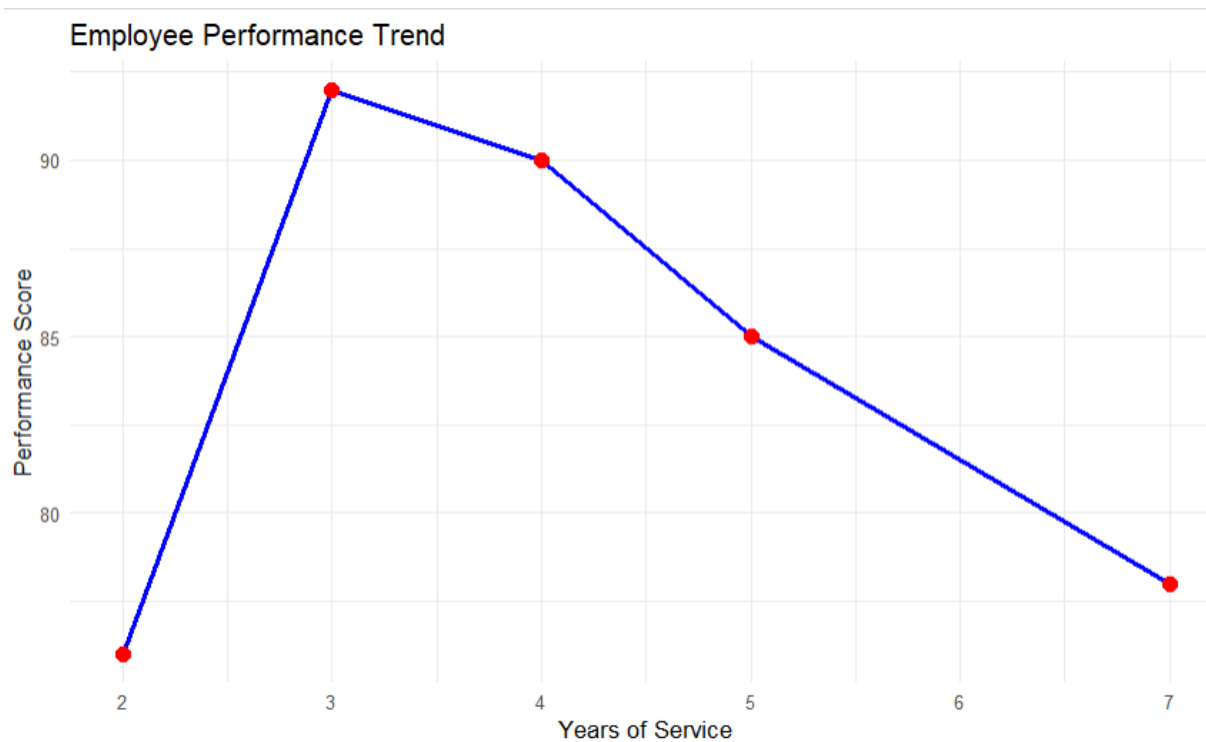
```
ggplot(employee,
```

```

aes(x = Years_of_Service,
    y = Performance_Score,
    group = 1)) +
geom_line(color = "blue",
    linewidth = 1) +
geom_point(color = "red",
    size = 3) +
labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
) +
theme_minimal()

```

Output



A line chart showing the trend between years of service and performance score.

Experiment 3: Tableau Dashboard

Step 1

Open **Tableau Public** or **Tableau Desktop**.

Step 2

Create an Excel file with the employee data (or export it from R) and connect it to Tableau.

Step 3: Create Worksheets

Worksheet 1 – Bar Chart

- Drag **Department** to **Columns**.
- Drag **Number of Records** to **Rows**.
- Set **Marks** to **Bar**.
- Drag **Department** to **Color**.
- Title: **Employee Distribution by Department**.

Worksheet 2 – Scatter Plot

- Create a new worksheet.
- Drag **Years_of_Service** to **Columns**.
- Drag **Performance_Score** to **Rows**.
- Set **Marks** to **Circle**.
- Drag **Department** to **Color**.
- Drag **Employee_ID** to **Label**.
- Title: **Years of Service vs Performance Score**.

Step 4: Create Dashboard

1. Click **Dashboard** → **New Dashboard**.
2. Add:
 - Bar Chart
 - Scatter Plot

Suggested Layout

```
+-----+
| Department Distribution | Years of Service vs Score |
+-----+
```

(Optional) Enable **Use as Filter** for interactive filtering.

Experiment 4: Employee Performance Table

View(employee)

OR

print(employee)

Output

Employee ID Department Years of Service Performance Score

1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

	Employee_ID	Department	Years_of_Service	Performance_Score	
1	1	Sales	5	85	
2	2	HR	3	92	
3	3	Marketing	7	78	
4	4	Sales	4	90	
5	5	HR	2	76	

4 columns

Employee_ID <dbl>

[1, 5] 0% NA

Department <chr>

3 unique 0% NA

Years_of_Service <dbl>

[2, 7] 0% NA

Performance_Score <dbl>

[76, 92] 0% NA

Showing 1 to 5 of 5 entries, 4 total columns

Set 24 Product Inventory Management

Dataset: Product Inventory

Product ID Product Name Quantity Available

1 Product A 250

2 Product B 175

3 Product C 300

4 Product D 200

5 Product E 220

1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.
3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.
4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

Step 1: Create the dataset

```
inventory <- data.frame(  
  Product_ID = c(1, 2, 3, 4, 5),  
  Product_Name = c(  
    "Product A", "Product B", "Product C",  
    "Product D", "Product E"  
  ),  
  Quantity_Available = c(250, 175, 300, 200, 220),  
  Category = c(  
    "Electronics", "Clothing", "Electronics",  
    "Clothing", "Electronics"  
  ),  
  Price = c(20, 15, 18, 25, 22)  
)
```

```
View(inventory)
```

1. Bar Chart — Quantity of Each Product

First install and load ggplot2:

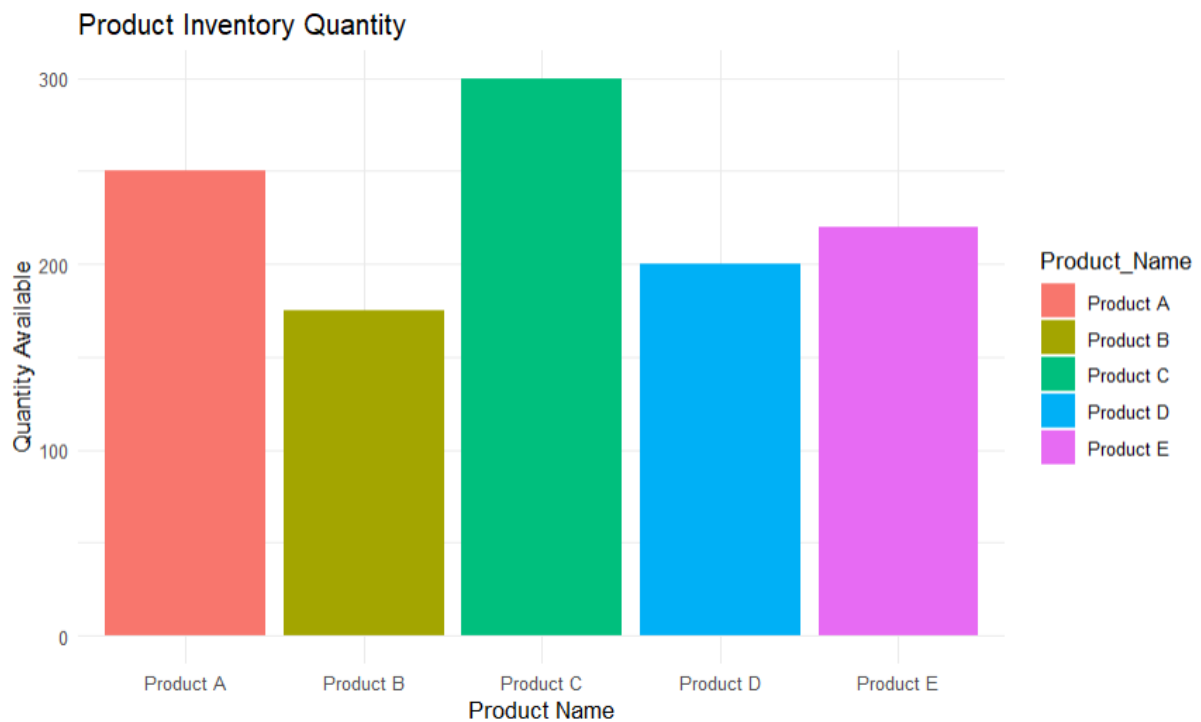
```
install.packages("ggplot2")
```

```
library(ggplot2)
```

Then run:

```
ggplot(inventory,
  aes(x = Product_Name,
      y = Quantity_Available,
      fill = Product_Name)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory Quantity",
    x = "Product Name",
    y = "Quantity Available"
  ) +
  theme_minimal()
```

Output



The chart compares the available quantity of Products A–E.

Observation: Product C has the highest inventory (300), while Product B has the lowest (175).

2. Stacked Bar Chart — Product Categories

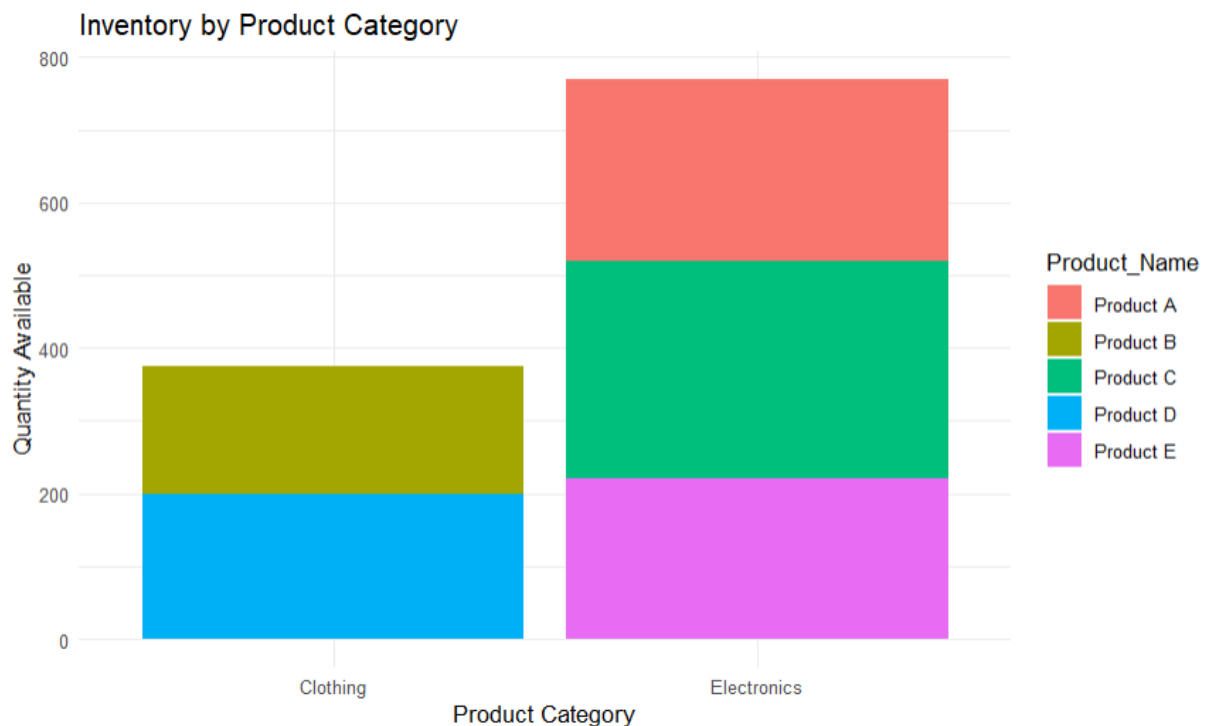
```
ggplot(inventory,
  aes(x = Category,
      y = Quantity_Available,
```

```

    fill = Product_Name)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Inventory by Product Category",
    x = "Product Category",
    y = "Quantity Available"
  ) +
  theme_minimal()

```

This shows the total inventory within each category, with individual products stacked inside the category.



3. Tableau Dashboard

For Tableau, use the same data.

Create the Bar Chart

1. Open **Tableau**.
2. Connect to your inventory data.
3. Drag **Product_Name** → Columns.
4. Drag **Quantity_Available** → Rows.
5. Select **Bar** under Marks.

6. Drag **Product_Name** → Color.

Create the Stacked Bar Chart

Create another worksheet:

1. Drag **Category** → Columns.
2. Drag **Quantity_Available** → Rows.
3. Drag **Product_Name** → Color.
4. Select **Bar**.

Create Dashboard

Go to:

Dashboard → New Dashboard

Drag both worksheets into the dashboard.

Suggested layout:

```
+-----+
| Product Inventory Bar Chart          |
+-----+
| Category-wise Stacked Bar Chart      |
+-----+
```

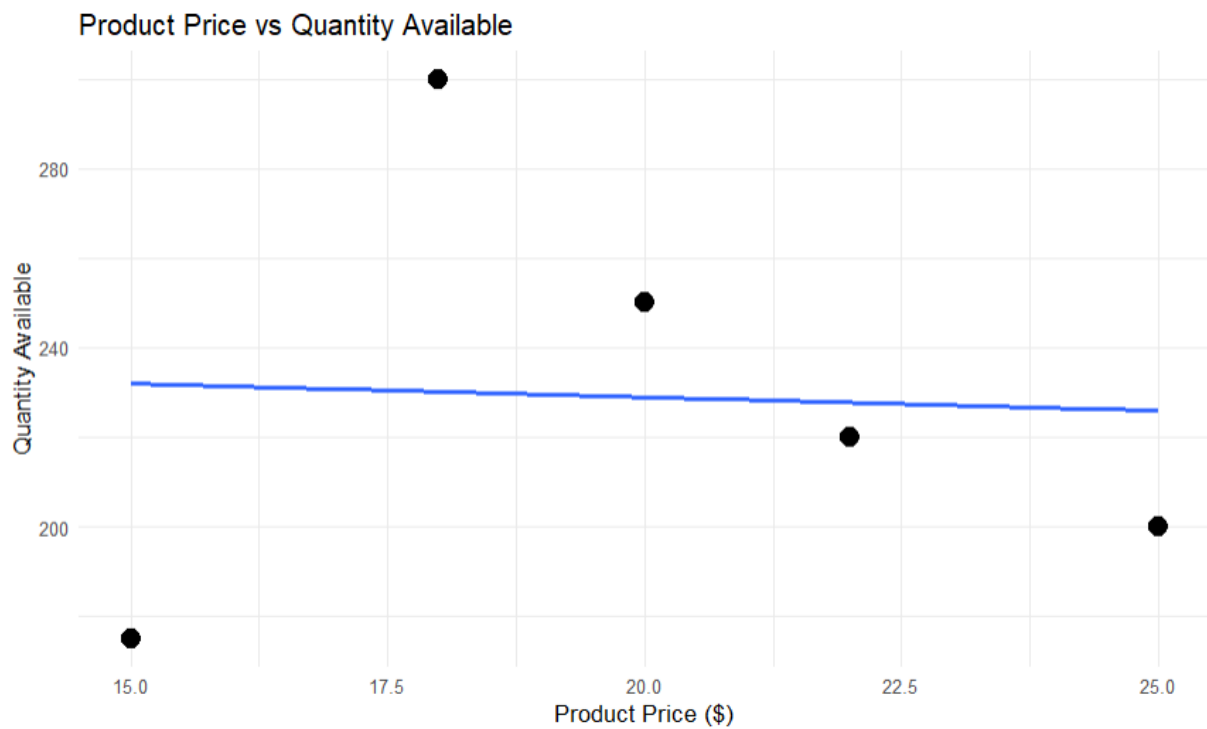
You can add **Category** and **Product Name** as filters to make it interactive.

4. Scatter Plot — Price vs Quantity

```
ggplot(inventory,
  aes(x = Price,
      y = Quantity_Available)) +
  geom_point(size = 4) +
  geom_smooth(
    method = "lm",
    se = FALSE
  ) +
  labs(
    title = "Product Price vs Quantity Available",
    x = "Product Price ($)",
    y = "Quantity Available"
```

) +

theme_minimal()



Interpretation

- Product C has a relatively low price (\$18) and the **highest quantity (300)**.
- Product D has the highest price (\$25) and a quantity of 200.
- With only five products, the relationship between price and quantity is **not strong enough to make a reliable conclusion**.

Set 25: Website Traffic Analysis

Dataset: Website Traffic

Date Page Views Click-through Rate

2023-01-01 1500 2.3%

2023-01-02 1600 2.7%

2023-01-03 1400 2.0%

2023-01-04 1650 2.4%

2023-01-05 1800 2.6%

1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.
2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.
3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.
4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

Step 1: Create the Dataset in R

```
traffic <- data.frame(  
  Date = as.Date(c(  
    "2023-01-01",  
    "2023-01-02",  
    "2023-01-03",  
    "2023-01-04",  
    "2023-01-05"  
  )),  
  Page_Views = c(1500, 1600, 1400, 1650, 1800),  
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)  
  
View(traffic)
```

1. Tableau — Line Chart for Page Views

This part is done in **Tableau**, not R.

Step 1

Open Tableau and connect to your website traffic data.

Step 2

Create a new worksheet.

Step 3

Drag:

- **Date** → Columns
- **Page_Views** → Rows

Step 4

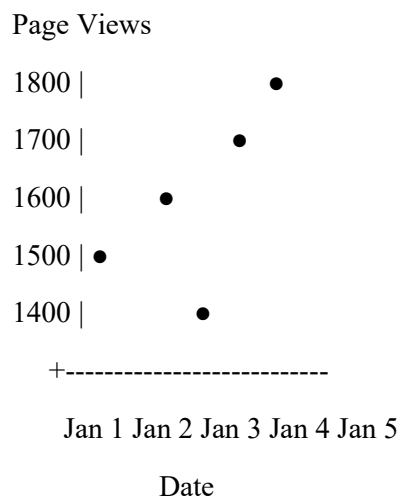
Select **Line** from the Marks card.

Step 5

Add the title:

Website Page Views Trend

Expected chart



2. R — Top N Days by Click-Through Rate

Here we'll take **Top 3 days**.

```
library(ggplot2)
```

```
top_n <- 3
```

```
top_days <- traffic[
  order(-traffic$Click_Through_Rate),
  ][1:top_n, ]
```

```
print(top_days)
```

Now create the bar chart:

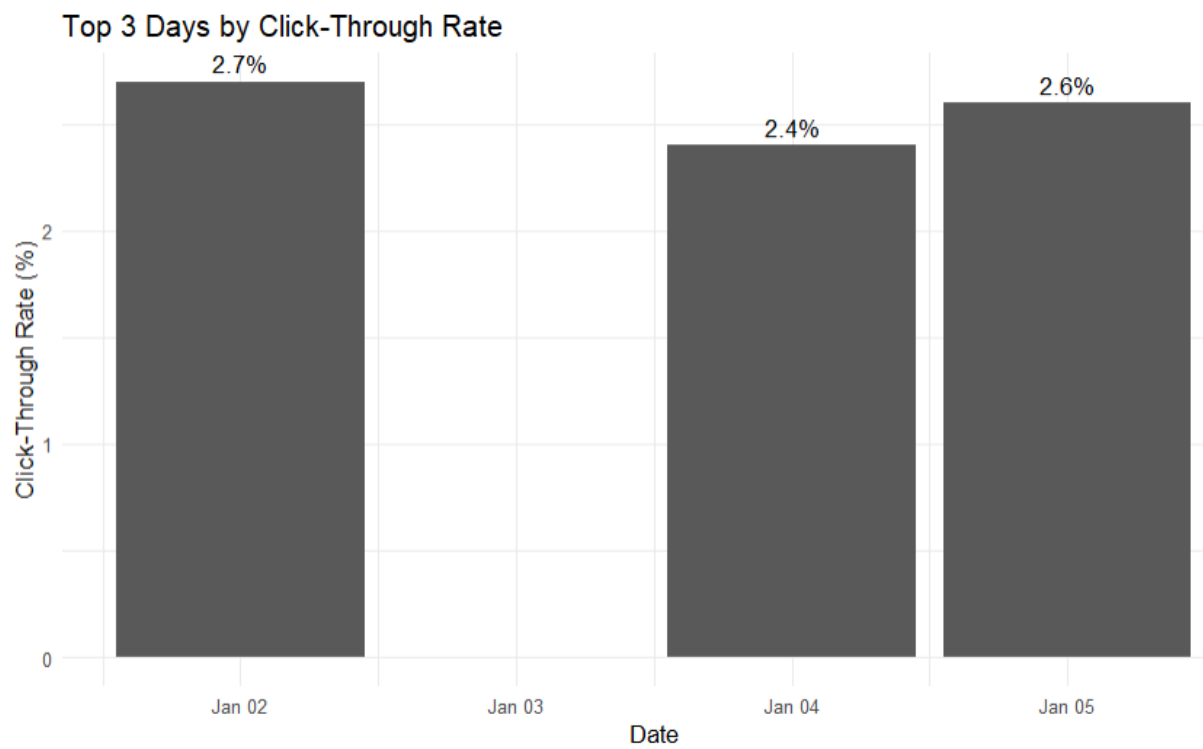
```
ggplot(top_days,
  aes(x = Date,
```

```

    y = Click_Through_Rate)) +
  geom_bar(stat = "identity") +
  geom_text(
    aes(label = paste0(Click_Through_Rate, "%")),
    vjust = -0.5
  ) +
  labs(
    title = "Top 3 Days by Click-Through Rate",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()

```

Expected result



The top 3 days are:

Date	Click-through Rate
------	--------------------

2023-01-02	2.7%
------------	------

2023-01-05	2.6%
------------	------

2023-01-04	2.4%
------------	------

3. R — Stacked Area Chart

Because the original dataset doesn't contain user interactions, we'll add sample interaction data.

```
interactions <- data.frame(  
  Date = as.Date(c(  
    "2023-01-01",  
    "2023-01-02",  
    "2023-01-03",  
    "2023-01-04",  
    "2023-01-05"  
  )),  
  Likes = c(120, 140, 110, 150, 170),  
  Shares = c(50, 60, 45, 70, 80),  
  Comments = c(30, 35, 25, 40, 45)  
)
```

View(interactions)

Convert the data to long format:

```
library(reshape2)  
interaction_long <- melt(  
  interactions,  
  id.vars = "Date",  
  variable.name = "Interaction",  
  value.name = "Count"  
)
```

Create the stacked area chart:

```
ggplot(  
  interaction_long,  
  aes(  
    x = Date,  
    y = Count,  
    fill = Interaction  
  )  
)
```

```
) +
  geom_area() +
  labs(
    title = "Website User Interactions",
    x = "Date",
    y = "Number of Interactions",
    fill = "Interaction Type"
  ) +
  theme_minimal()
```

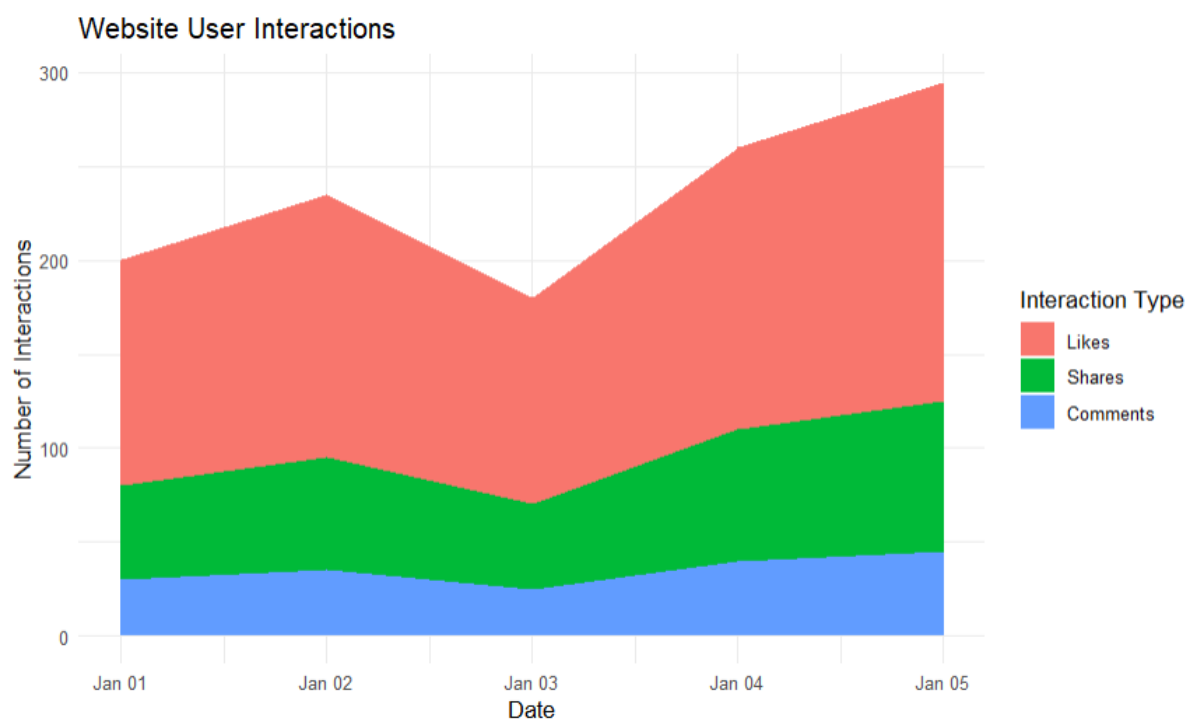


Chart contains

- Likes
- Shares
- Comments

4. Tableau — Interactive Map + Page Views by Source

The original dataset does not contain **traffic sources** or **geographic locations**, so create a second dataset with sample values.

For example:

Source	City	Page Views
Google	New York	2500
Facebook	London	1800
Instagram	Mumbai	1500
YouTube	Sydney	1200
Direct	Toronto	1000

In Tableau

Create a data table containing:

Source

City

Page Views

Worksheet 1 — Interactive Map

1. Drag **City** into the worksheet.
2. Tableau should recognize it as a geographic field.
3. Select **Map**.
4. Drag **Page Views** → Size.
5. Drag **Source** → Color.
6. Drag **Source** → Label.

You now have a map showing traffic sources geographically.

Worksheet 2 — Page Views by Source

1. Create a new worksheet.
2. Drag **Source** → Columns.
3. Drag **Page Views** → Rows.
4. Select **Bar**.
5. Add the title:

Page Views by Traffic Source

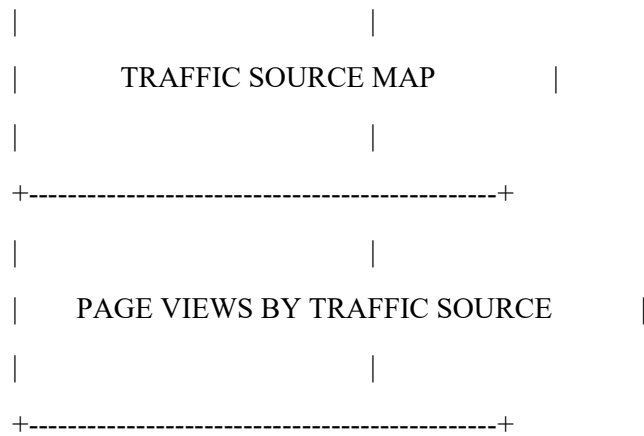
Create Dashboard

Go to:

Dashboard → **New Dashboard**

Drag both worksheets onto the dashboard:

+-----+



To make it interactive:

1. Click the map.
2. Select **Use as Filter**.
3. Click a source/location on the map.
4. The bar chart will update accordingly.

Final requirements

1. Page Views Line Chart
2. Top N Click-through Rate
3. Likes/Shares/Comments Area Chart
4. Traffic Source Map + Bar Chart

SET 26: Dataset: Student_Mini_Data.csv

Student_ID Gender Study_Hours Attendance Math_Score Science_Score
Exam_Date

S01 Male 2.0 78 62 65 2025-01-10

S02 Female 3.5 90 80 85 2025-01-10

S03 Male 1.5 70 55 58 2025-02-12

S04 Female 4.0 95 90 92 2025-02-12

S05 Male 2.8 85 72 74 2025-03-15

S06 Female 3.0 92 82 86 2025-03-15

1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.
2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.
3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.
4. Import Student_Mini_Data.csv into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

1. Histogram and Boxplot

R Code

```
library(ggplot2)

student <- data.frame(
  Student_ID = c("S01", "S02", "S03", "S04", "S05", "S06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Study_Hours = c(2, 3.5, 1.5, 4, 2.8, 3),
  Attendance = c(78, 90, 70, 95, 85, 92),
  Math_Score = c(62, 80, 55, 90, 72, 82),
  Science_Score = c(65, 85, 58, 92, 74, 86),
  Exam_Date = c("2025-01-10", "2025-01-10",
                "2025-02-12", "2025-02-12",
                "2025-03-15", "2025-03-15")
)

student$Exam_Date <- as.Date(student$Exam_Date)

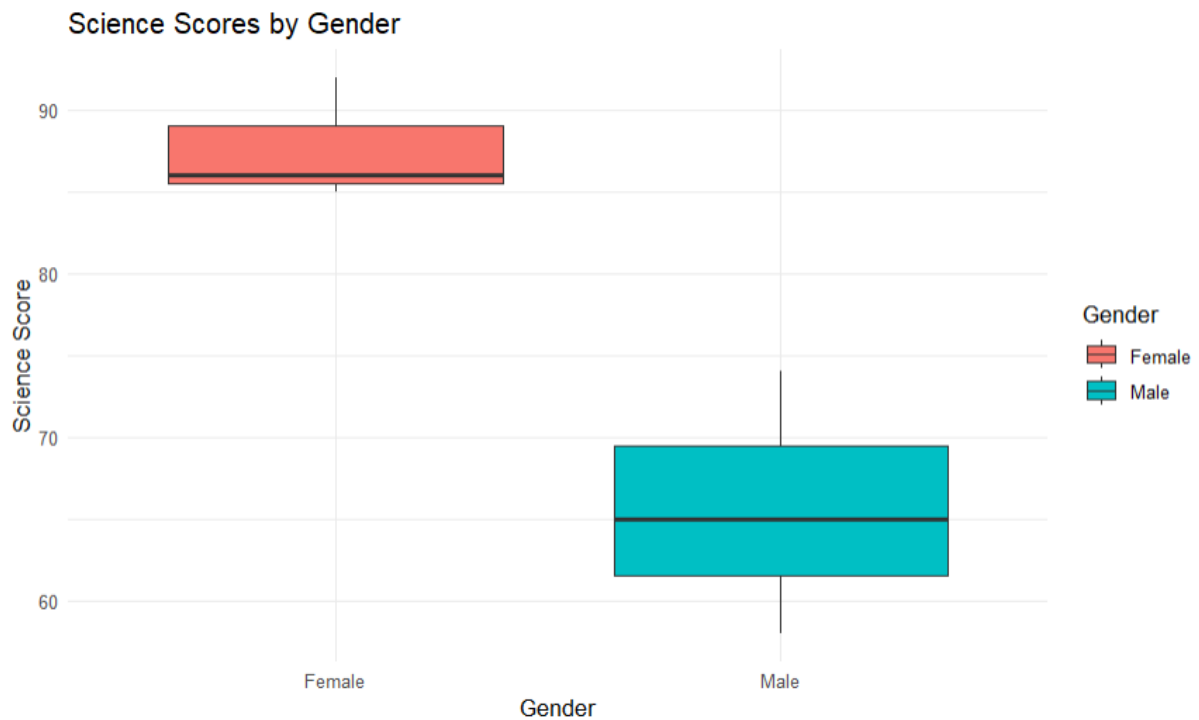
# Histogram
ggplot(student, aes(x=Math_Score)) +
  geom_histogram(binwidth=5, fill="skyblue", color="black") +
```

```
labs(
  title="Distribution of Math Scores",
  x="Math Score",
  y="Number of Students"
) +
theme_minimal()
```

Boxplot

```
ggplot(student, aes(x=Gender, y=Science_Score, fill=Gender)) +
  geom_boxplot() +
  labs(
    title="Science Scores by Gender",
    x="Gender",
    y="Science Score"
  ) +
  theme_minimal()
```





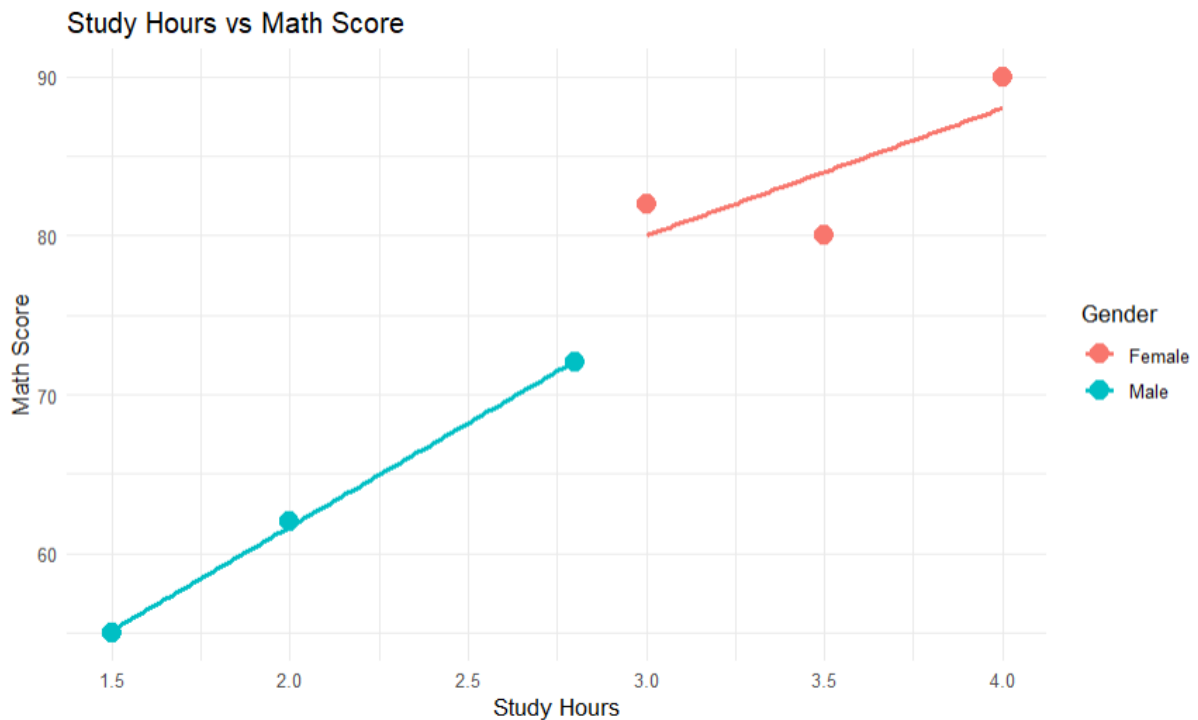
Interpretation

The histogram shows the distribution of Math scores. The boxplot shows that the female students generally have higher Science scores than the male students.

2. Scatter Plot – Study Hours vs Math Score

R Code

```
ggplot(student,
  aes(x=Study_Hours,
    y=Math_Score,
    color=Gender)) +
  geom_point(size=4) +
  geom_smooth(method="lm", se=FALSE) +
  labs(
    title="Study Hours vs Math Score",
    x="Study Hours",
    y="Math Score",
    color="Gender"
  ) +
  theme_minimal()
```



Interpretation

The scatter plot shows a **positive relationship** between study hours and Math scores. Students who spend more time studying generally obtain higher Math scores.

3. Monthly Average Math Score and Moving Average

R Code

```
student$Month <- format(student$Exam_Date, "%Y-%m")
```

```
monthly_avg <- aggregate(
```

```
  Math_Score ~ Month,
```

```
  data=student,
```

```
  FUN=mean
```

```
)
```

```
names(monthly_avg)[2] <- "Average_Math"
```

```
# Moving average
```

```
monthly_avg$Moving_Average <- c(
```

```
  NA,
```

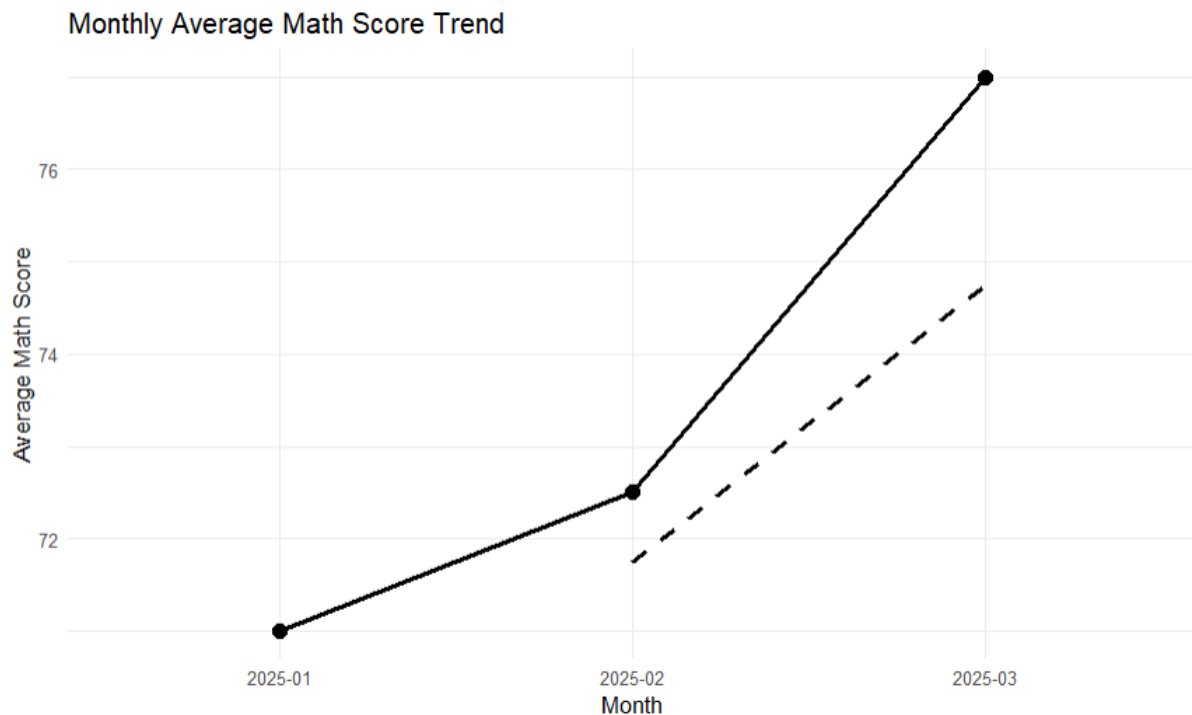
```
mean(monthly_avg$Average_Math[1:2]),  
mean(monthly_avg$Average_Math[2:3])  
)
```

```
print(monthly_avg)
```

```
# Line chart
```

```
ggplot(monthly_avg, aes(x=Month)) +  
  geom_line(  
    aes(y=Average_Math, group=1),  
    linewidth=1  
  ) +  
  geom_point(  
    aes(y=Average_Math),  
    size=3  
  ) +  
  geom_line(  
    aes(y=Moving_Average, group=1),  
    linewidth=1,  
    linetype="dashed"  
  ) +  
  labs(  
    title="Monthly Average Math Score Trend",  
    x="Month",  
    y="Average Math Score"  
  ) +  
  theme_minimal()
```

Output



Month	Average_Math
2025-01	71.0
2025-02	72.5
2025-03	77.0

Interpretation

The average Math score increases from **71 to 77**, showing an **increasing performance trend** from January to March.

4. Tableau Dashboard

Step 1 – Import Dataset

Import Student_Mini_Data.csv into Tableau.

Step 2 – Bar Chart

Use:

Gender → Columns

Math_Score → Rows

Change SUM(Math_Score) to **Average**.

Title:

Average Math Score by Gender

Step 3 – Line Chart

Use:

Exam_Date → Columns

Math_Score → Rows

Select **Line Chart**.

Title:

Math Score Trend Over Time

Step 4 – Filter

Drag:

Exam_Date → Filters

Select **Show Filter**.

Step 5 – Dashboard

Create **New Dashboard** and add:

- Average Math Score by Gender
- Math Score Trend Over Time
- Exam Date filter

Result: An interactive dashboard for exploring student performance by gender and examination date.

SET 27.

Patient Health Risk Analysis

Patient\ _ID Age BMI BP Cholesterol

P1 25 22 120 180

P2 40 28 135 210

P3 55 30 145 240

P4 35 26 130 200

P5 60 32 150 260

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.
2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

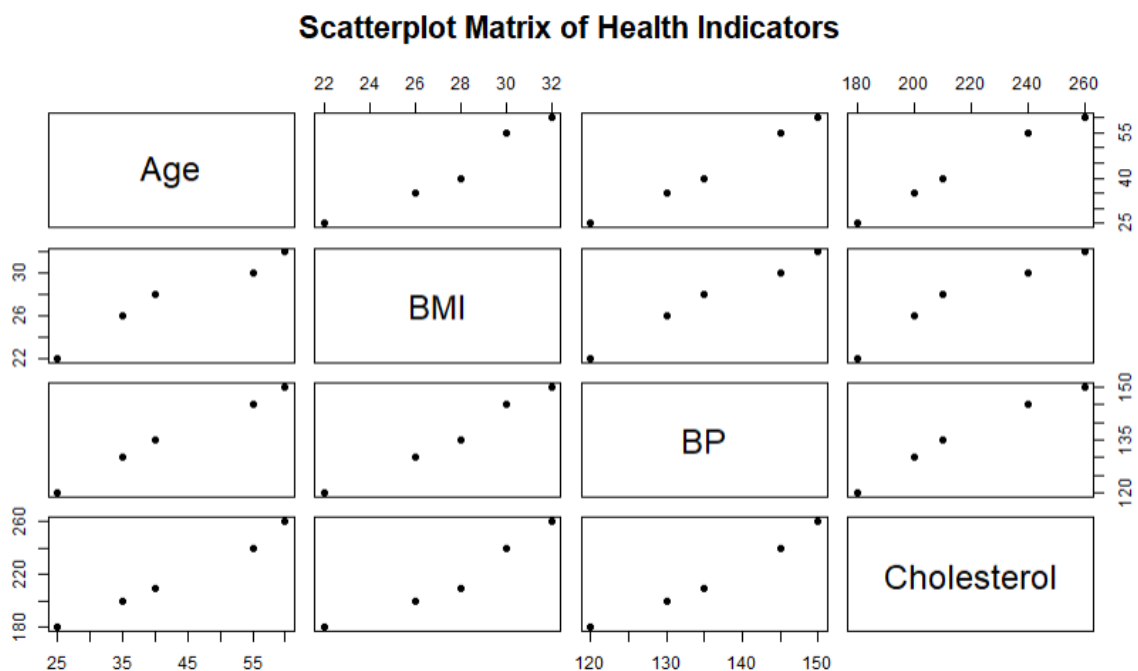
3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

1. Scatterplot Matrix

R Code

```
pairs(  
  patient[, c("Age", "BMI", "BP", "Cholesterol")],  
  main = "Scatterplot Matrix of Health Indicators",  
  pch = 19  
)
```



Interpretation

The scatterplot matrix shows positive relationships between **Age**, **BMI**, **BP**, and **Cholesterol**. Older patients generally have higher BMI, blood pressure, and cholesterol levels. Patient P5 shows particularly high-risk values.

2. Q-Q Plot and ECDF for Cholesterol

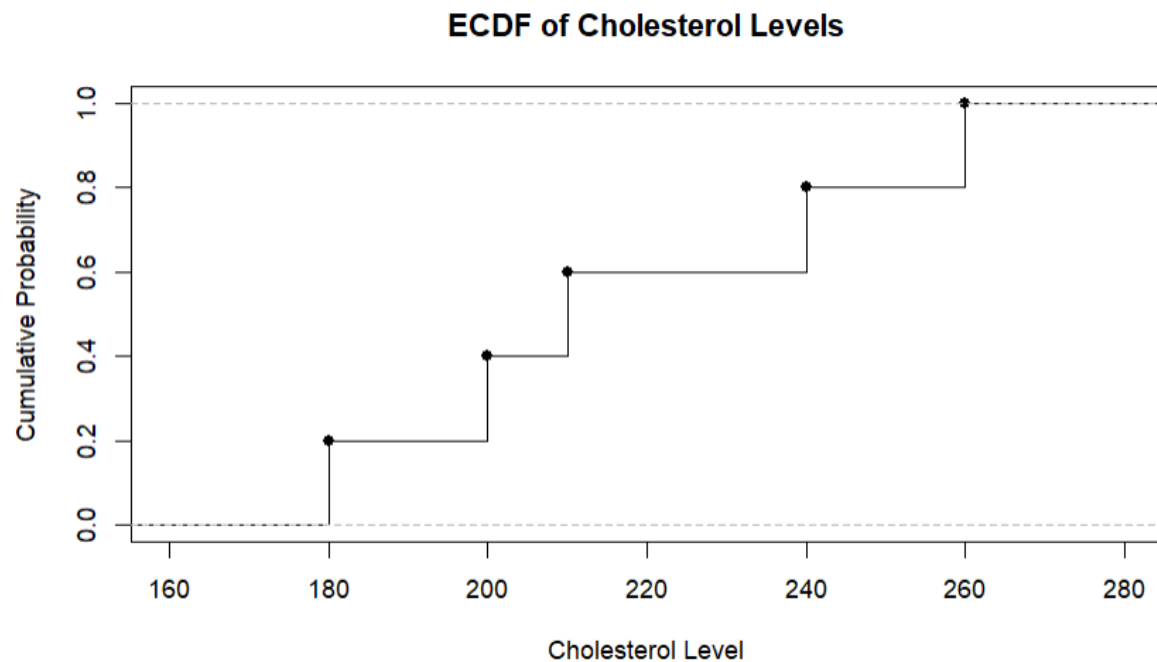
Q-Q Plot

```
qqnorm(  
  patient$Cholesterol,  
  main = "Q-Q Plot of Cholesterol"  
)
```

```
qqline(  
  patient$Cholesterol,  
  col = "red"  
)
```

ECDF

```
plot(  
  ecdf(patient$Cholesterol),  
  main = "ECDF of Cholesterol Levels",  
  xlab = "Cholesterol",  
  ylab = "Cumulative Probability"  
)
```



Interpretation

The Q-Q plot is used to compare cholesterol values with a normal distribution. Because the dataset contains only five observations, normality cannot be reliably determined. The ECDF shows the cumulative distribution of cholesterol values from **180 to 260**.

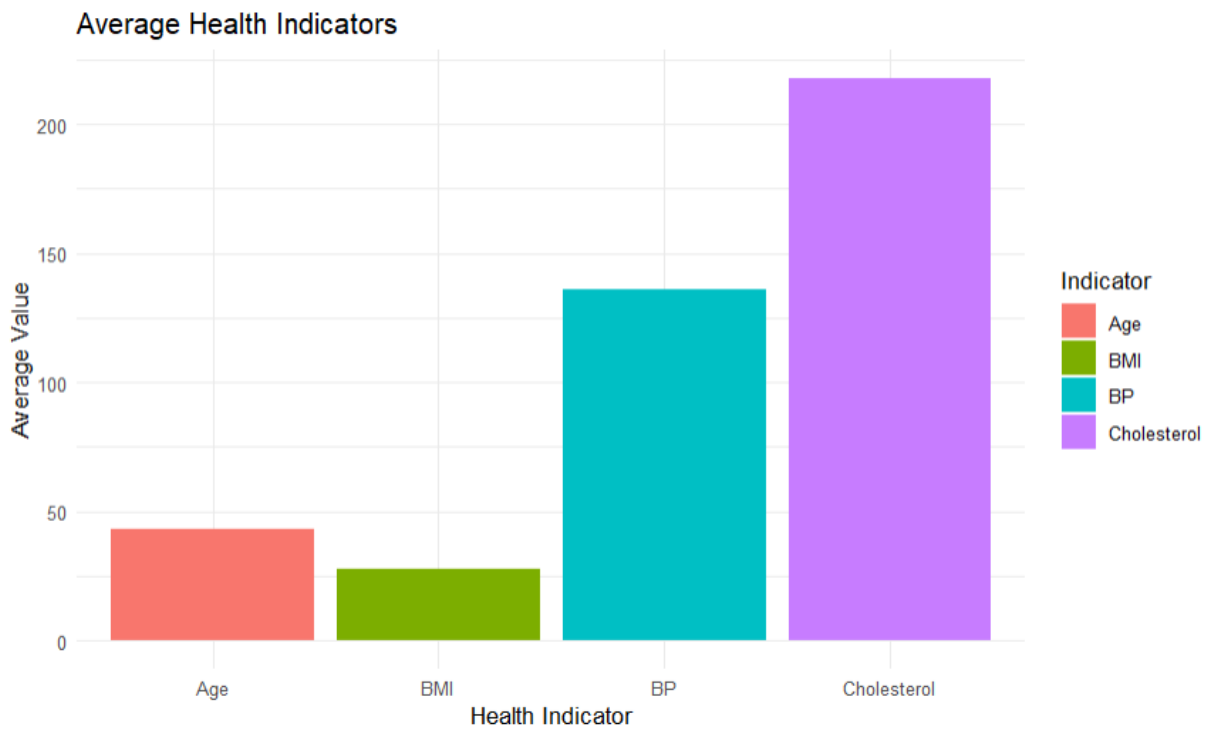
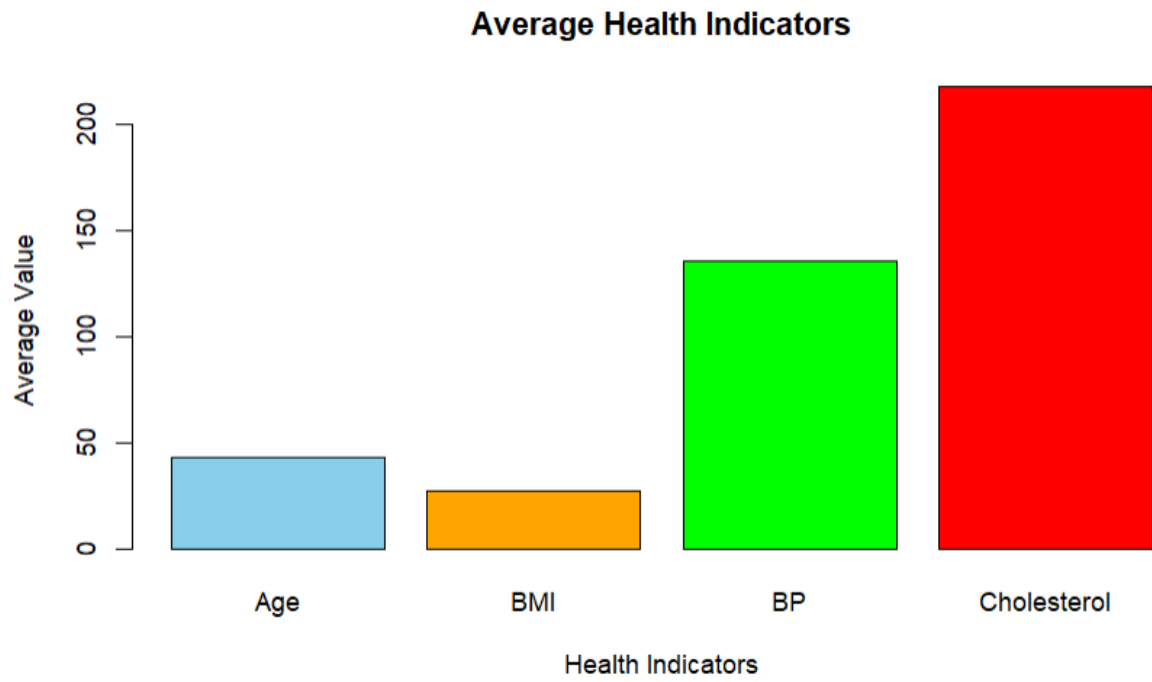
3. Average Health Indicators Bar Chart

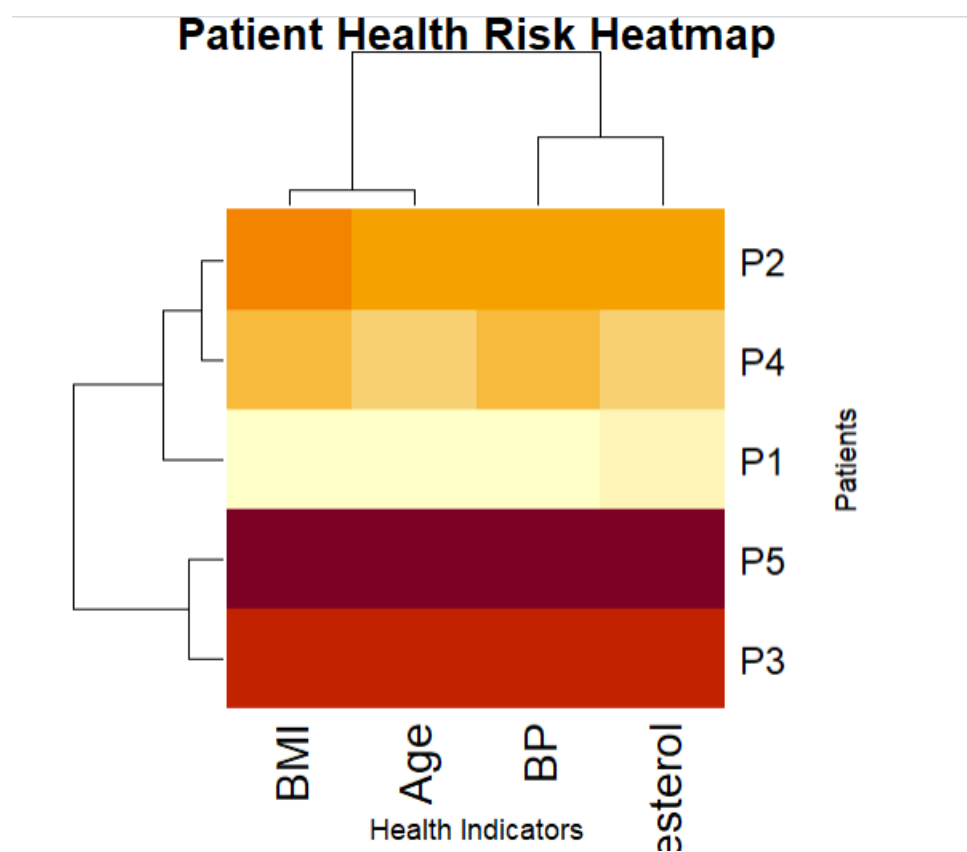
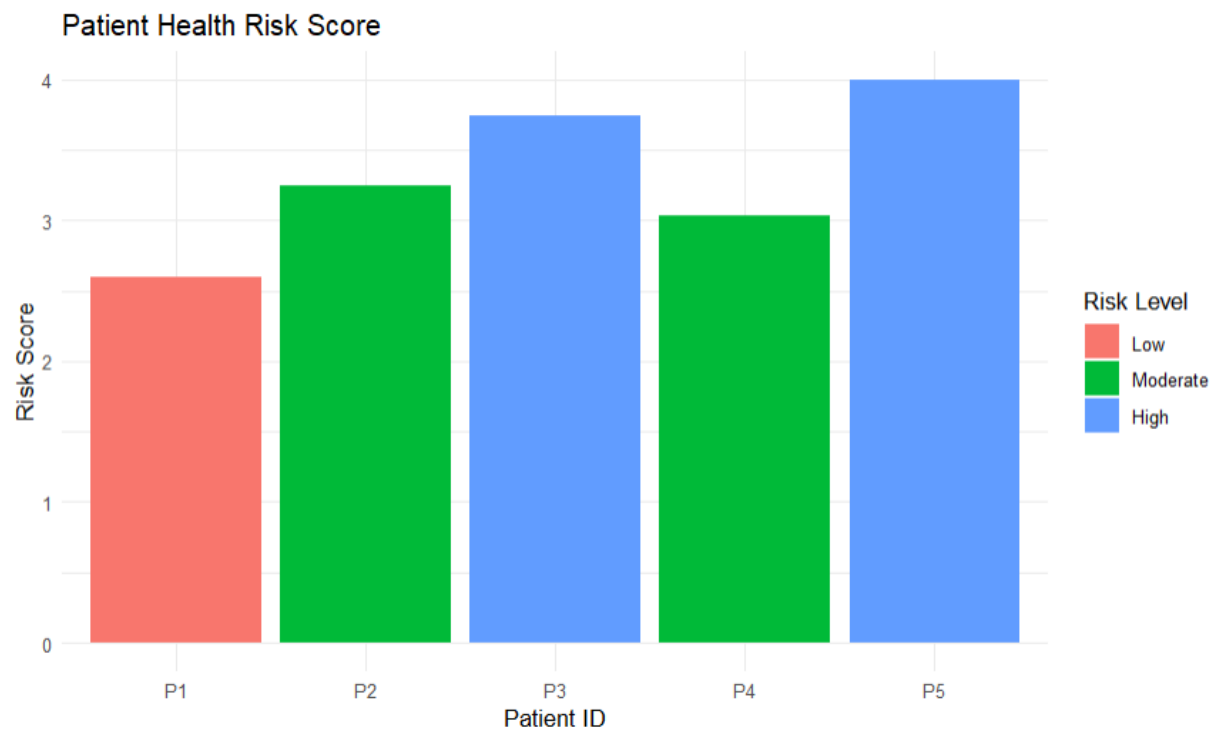
R Code

```
averages <- c(  
  Age = mean(patient$Age),  
  BMI = mean(patient$BMI),  
  BP = mean(patient$BP),  
  Cholesterol = mean(patient$Cholesterol)  
)  
barplot(  
  averages,  
  col = c("skyblue", "orange", "green", "red"),  
  main = "Average Health Indicators",  
  xlab = "Health Indicators",  
  ylab = "Average Value"  
)
```

Average Values

```
Age      = 43  
BMI      = 27.6  
BP       = 136  
Cholesterol = 218
```





Interpretation

The average values provide an overall view of the patient group. Higher BMI, BP, and cholesterol can indicate increased health risk, although proper clinical risk assessment requires medical reference ranges and additional patient information.

4. Tableau Dashboard

Step 1 – Create Heatmap

Import the patient dataset into Tableau.

Drag:

Age → Columns

Patient_ID → Rows

Drag:

Cholesterol → Color

Also add:

BMI → Label

BP → Label

Cholesterol → Label

Age → Label

Choose **Square** marks.

Step 2 – Add Age Filter

Drag:

Age → Filters

Select **Show Filter**.

Step 3 – Add Tooltip

Click **Tooltip** and include:

Patient ID

Age

BMI

BP

Cholesterol

Step 4 – Create Dashboard

Create:

Dashboard → **New Dashboard**

Add the heatmap and Age filter.

Risk Analysis

Based on the given values, **P5** has the highest combination of age, BMI, BP, and cholesterol and therefore appears to have the highest risk profile in this small dataset. **P3** also shows elevated values and may require attention.

The dashboard allows healthcare professionals to quickly filter patients by age and identify patients with multiple elevated indicators for further assessment.